

Ian Leung

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Education

University of Waterloo

2025 – 2030

Bachelor of Computer Science and Finance - Double Major

GPA: 3.96/4.0

- Functional Programs (Advanced), Algorithms (Advanced), Linear Algebra, Data Analytics and Financial Markets
- President's Scholarship of Distinction (\$5000)

Technical Skills

Languages Python, Java, C++, C#, C, Rust, Kotlin, JavaScript, SQL, HTML, CSS, Racket, Bash
Technologies PyTorch, Firebase, MongoDB, Docker, Render, Modal, Qt, OpenCV, PostgreSQL, SQLite
Developer Tools Linux, Git, GitLab, GitHub, Kaggle, Colab, Jupyter, Unity, Android Studio, Copilot

Work Experience

Roblox

San Mateo, CA

Incoming Software Engineering Intern | C++

Jun 2026 – Aug 2026

- Incoming Summer 2026

Sunnybrook Research Institute

Toronto, ON

Software Research Intern | Python

Jul 2025 – Aug 2025

- Built an event tracing tool to simulate medical professionals' interactions with **Focused Ultrasound** 🧠 software
- Reduced software release test times by **91%** by creating an automated testing framework with **PyQt** and **Pydantic**
- Implemented a logging API using **Difflib** and **Loguru** to validate test results against baselines, identifying **5+** bugs

Data Science Club

Waterloo, ON

Software Developer & Hackathon Organizer | TypeScript, HTML/CSS, Supabase

Sep 2025 – Present

- Maintaining www.uwdatascience.ca - UWDC's primary website and associated apps for events and internal tooling
- Managed food and logistics for 350+ hackathon attendees at **CxC** 🧠 - Canada's largest student-run AI hackathon
- Negotiated contracts with **15+** food vendors to deliver meals and snacks within a **\$10,000** budget

Waterloo Reality Labs

Waterloo, ON

Software Developer | Python

Sep 2025 – Present

- Developing **Universal Gestures** 🧠 - an open-source hand gesture recognition library for VR/AR applications
- Implemented a real-time **PyTorch** training pipeline enabling users to define custom hand gestures via **Unity Plugin**

Projects

Audio Source Separation Transformer Model | PyTorch, Asteroid, Nsound, Aiotone

- Expanding on **Meta AI's Hybrid Transformer** 🧠 models to separate audio components from universal audio sources
- Synthesizing 1000+ hours of custom audio data using **Nsound** and **Aiotone** for model training and evaluation
- Architecting a mobile app deployment for real-time audio identification and isolation in global environments

🧠 Pharmagotchi: AI Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built an Android app using **OpenRouter LLMs** to provide AI-driven health insights and medication guidance to users
- Automated medication reminders, prescription tracking, and health report generation for healthcare providers

🧠 PeerAssist - Full-Stack Peer Reviewing | Java, MongoDB, Apache PDFBox, Maven

- Designed and developed a full-stack Java application for 5+ students to share, review and grade the works of others
- Integrated **MongoDB** for data storage and **Apache PDFBox** for PDF manipulation, enabling seamless document handling and retrieval across devices

🧠 Trust No Ghost - 3D Maze Exploration Game | Unity, C#, Itch.io

- Shipped Trust No Ghost 🧠 for **60+** users at the UW Game Jam Fall 2025, placing in the **top 10** most rated games
- Utilized Unity and C# to develop expanding level designs, AI pathfinding algorithms, and dynamic moving mechanics

Other Projects

- 🧠 **Chess Engine:** MCTS Chess Engine Deployed for Chesshacks 🧠. Built with PyTorch, Modal, Hugging Face
- 🧠 **K-Pop Idol Recognizer:** LBPH Facial Detection and Recognition. Built with OpenCV, Kaggle, Pandas, NumPy
- 🧠 **AI Name Forecasting:** Predicting future outlook using AI, based on Implicit Egotism. Built with OpenAI API